

can solve the problem of ‘double spending’ without the need for a central authority or figure. All transactions are transparent and consensual, ensuring that all parties involved have equal power and access. This also solves the problem of inflation that happens commonly with cash based currencies.

Blockchain technology, as it is now known, is being used to revolutionise the way many sectors operate. Its implementation in cryptocurrencies is the most essential of these. Bitcoin, the most popular cryptocurrency in this field, is mined and collected all around the world. El Salvador has recently adopted Bitcoin as legal tender, and many more countries appear to be following suit.

Medical record systems, voting mechanisms, tax collection and a variety of other initiatives that require transparency, consensus and security are increasingly incorporating blockchain technology.

How does blockchain work?

Let’s now spend a few minutes attempting to figure out how the technology works. Wrapping your head around blockchain is a piece of cake if you’ve studied data structures and/or are familiar with the ‘linked list’ data structure. If you don’t understand something, you may need to read it again.

The notion of blockchain, like other breakthrough technologies, is really simple and straightforward. As seen in Figure 2, each block consists of three major parts. We have the data we want to store, as well as a hash code that uniquely identifies the block, and the hash code of the block before it in the chain. The data you want to save is determined by the technology for which you’re trying to use the blockchain. For example, if blockchain is used to hold data for a cryptocurrency, each block would have information such as the sender’s ID, receiver’s ID, amount transacted, and so on. A new block is created each time a transaction occurs.

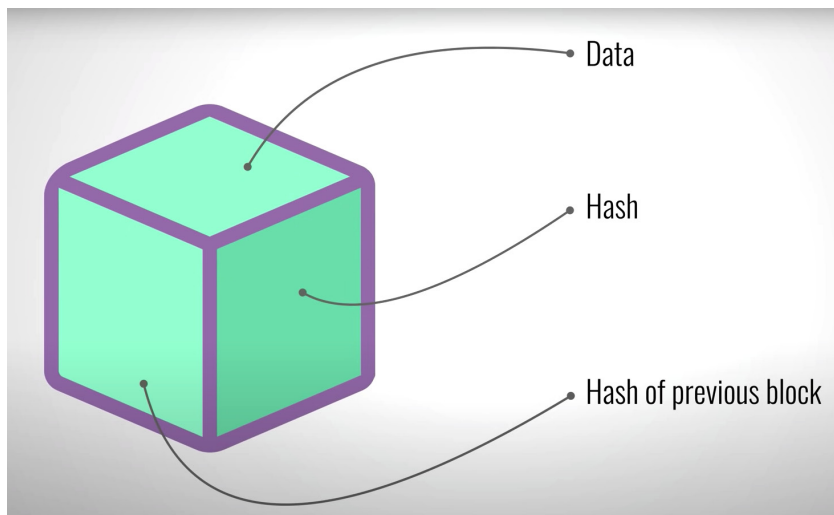


Figure 2: Elements of a block in blockchain (Courtesy: Simply Explained)

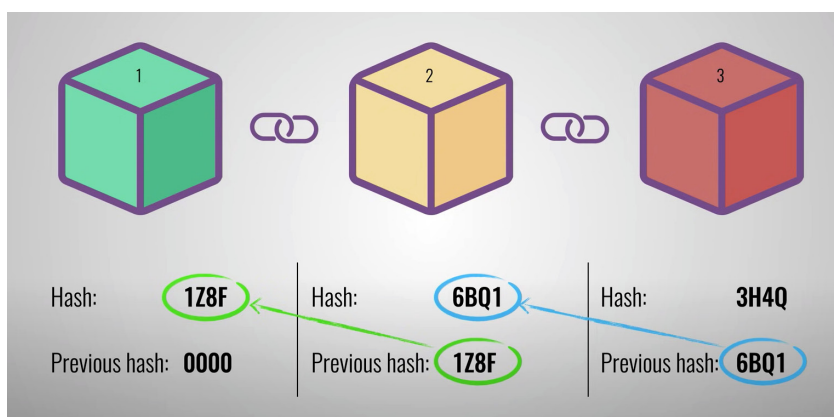


Figure 3: Function of a blockchain (Courtesy: Simply Explained)

So, why do we refer to blockchains as ‘immutable’? Figure 3 displays the formation of a blockchain and the contents of each block. When you add another block to the chain in Figure 3, a new hash is generated in addition to the previous block’s hash.

The important thing to know is that every time you edit the data within a block, the hash value of that block is updated. If you tamper with or change the data in an existing block, the hash code of that block will change as well. This hash code is unrecognisable to its successor, who has stored the prior block’s original hash, resulting in the blockchain being broken. The chain is broken when any block is tampered with (see Figure 4).

Blockchains are believed to be immutable for the reason that data remains intact once it is saved. So, given this constraint, how do blockchains handle adding new blocks? This issue is addressed with a notion known as ‘proof of work’. In essence, the process of adding blocks is slow and a block is only added after the transaction has been verified by enough individuals in the network. A proof of work is established once the transaction is verified, and the block is added to the chain. The time it takes to complete this operation varies depending on the blockchain implementation. In the case of Bitcoin, it takes roughly 10 minutes.